Flow Around Airfoil: Classical Methods, CFD and Physics-Informed Neural Networks

Mohammadreza Sotoudeh · Fluid Mechanics 2 course project, Department of Mechanical Engineering, Sharif University of Technology · September 2025 – February 2026

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Abstract

This project studies steady, incompressible flow around NACA-series airfoils (NACA 6409, NACA 0024 and a family of symmetric NACA 00xx sections) using three complementary approaches: (1) the classical source panel method implemented in MATLAB, (2) high-fidelity CFD in ANSYS Fluent, and (3) Physics-Informed Neural Networks (PINNs) implemented in Python. The goal is to compare the three methods in terms of accuracy, computational cost and their ability to reproduce key aerodynamic quantities: pressure coefficient (Cp), lift and drag coefficients (Cl, Cd), velocity and pressure contours, and streamlines.

The panel method gives a fast potential-flow baseline. Fluent resolves viscous effects and near-stall behavior and agrees well with experiments for lift, but shows a serious drag-prediction problem at high angles of attack. The PINNs reproduce the flow field without any mesh, although the viscous (Navier–Stokes) PINN was run only at a low Reynolds number and its force coefficients differ strongly from the reference data.

1. Introduction

Airfoils are among the most fundamental components of aerodynamic systems such as aircraft wings, turbomachinery blades and energy converters. Accurate prediction of the flow around an airfoil is essential for optimizing aerodynamic performance and reducing drag. Airfoil flow is usually analyzed with numerical methods or experiments; with the emergence of machine learning, data-driven and physics-based approaches that solve the governing equations without a mesh have become possible.

In this work three approaches are applied to the same family of problems:

Approach	Tool	Role
Classical	Source panel method (MATLAB)	Fast potential-flow baseline
Grid-based CFD	ANSYS Fluent	High-fidelity reference (RANS)
Mesh-free ML	PINNs (Python/PyTorch)	Solve the governing equations with physics residuals as soft constraints

The project is organized into five phases:

Phase	Content	Airfoil	Tool
1	CFD, Cl/Cd, qualitative and quantitative mesh independence	NACA 6409	Fluent
2	Steady Navier–Stokes PINN	NACA 6409	Python
3 (part 1)	CFD at several angles of attack, lift/drag/L-D curves, Cp, comparison with experiment	NACA 6409	Fluent
3 (part 2)	Inviscid CFD at 5°	NACA 0024	Fluent
4	Potential-flow PINN	NACA 0024	Python
5	Source panel method, thickness effect, panel independence, validation	NACA 00xx	MATLAB

2. Airfoil Geometry and Flow Assumptions

NACA 6409 is a four-digit cambered airfoil with maximum camber of 6 % of the chord located at 40 % chord, and a maximum thickness of 9 % of the chord. **NACA 0024** is a symmetric airfoil with 24 % maximum thickness.

Common fluid assumptions: steady, incompressible, two-dimensional flow of a Newtonian fluid (air, $\rho = 1.225$ kg/m³, $\mu = 1.85\times10^{-5}$ Pa·s for the viscous cases).

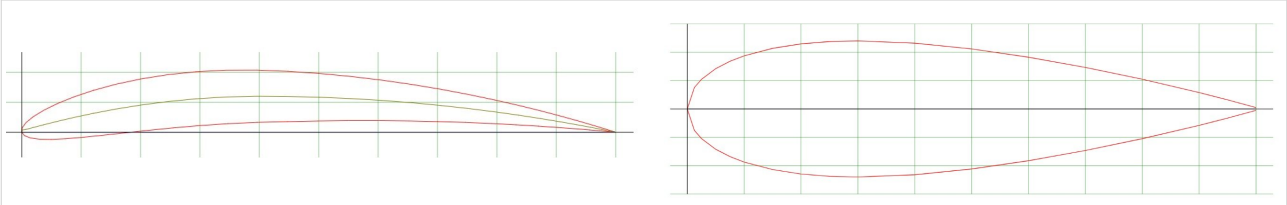


Figure 1. *Airfoil geometries: (a) NACA 6409, (b) NACA 0024.*

3. Governing Equations

For viscous flow (CFD and the Navier–Stokes PINN) the continuity and momentum equations are solved:

- Continuity: $\partial u/\partial x + \partial v/\partial y = 0$
- x-momentum: $\rho(u\,\partial u/\partial x + v\,\partial u/\partial y) = -\partial p/\partial x + \mu(\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2)$
- y-momentum: $\rho(u\,\partial v/\partial x + v\,\partial v/\partial y) = -\partial p/\partial y + \mu(\partial^2 v/\partial x^2 + \partial^2 v/\partial y^2)$

In Fluent, turbulence is modeled with the Standard k–ε model; the Reynolds number is $Re = VL/\nu$.

For potential (inviscid, irrotational) flow, used by the panel method and the potential-flow PINN:

- Continuity: $\partial u/\partial x + \partial v/\partial y = 0$
- Irrotationality: $\partial v/\partial x - \partial u/\partial y = 0$
- Laplace equation for the velocity potential: $\nabla^2\varphi = \partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 = 0$

4. Methodology

4.1 CFD in ANSYS Fluent

- **Mesh:** structured C-type meshes around the airfoil with several refinement levels.
- **Turbulence model:** Standard k–ε with Enhanced Wall Treatment (the standard wall approach is appropriate for y^+ between about 30 and 300, and enhanced wall treatment for lower y^+).
- **Reynolds number:** 2×10^6 (freestream speed about 30.21 m/s) and 3×10^6 (about 45.31 m/s).
- **Boundary conditions:** uniform velocity inlet, zero gauge pressure at the outlet, no-slip condition on the airfoil surface.
- **Angles of attack:** 0° and 5° for the first study; −5°, 0° and 15° (near stall) for the angle-of-attack study, with additional points up to 20° for the lift curve.
- **NACA 0024 case:** inviscid flow ($\mu = 0$) at 5°, so no turbulence model is active and y^+ has no physical meaning.

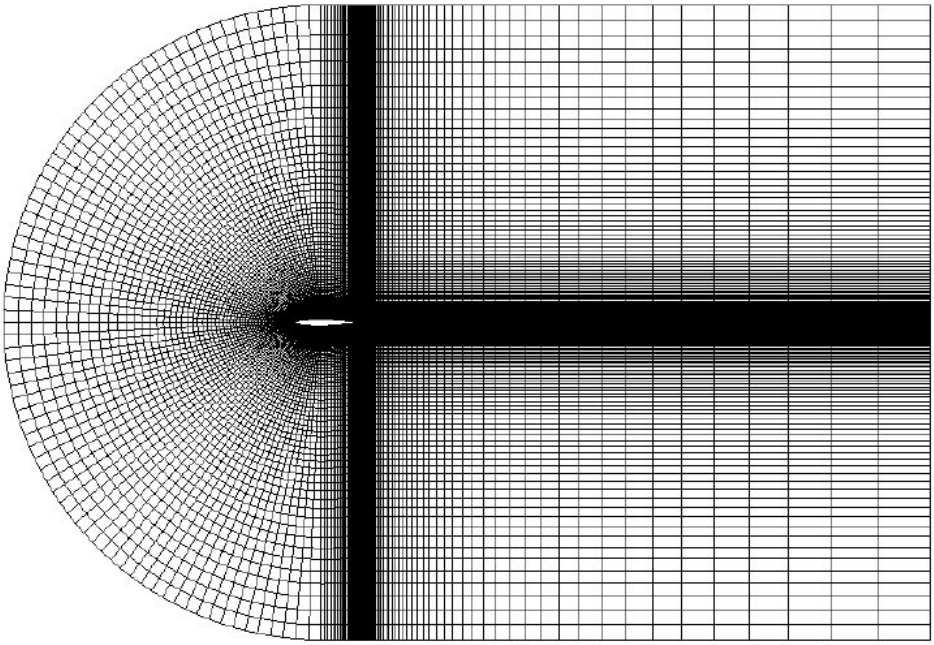


Figure 2. *C-type mesh domain used in the Fluent simulations.*

4.2 Physics-Informed Neural Networks

PINNs approximate the solution with a fully connected neural network and train it by minimizing the residuals of the governing equations and the boundary conditions, so no mesh is required.

Navier–Stokes PINN (NACA 6409)

- Input: coordinates (x, y); output: velocity components u, v and pressure p.
- Architecture: multilayer perceptron with 6 hidden layers of 100 neurons and tanh activation (smooth and differentiable, as needed for computing physical gradients).
- Loss: weighted sum of the boundary-condition loss (mean-squared error of u, v, p at the boundaries) and the PDE loss (continuity and two momentum residuals).
- Optimization: Adam with mini-batches and a learning-rate scheduler, followed by L-BFGS phases for refinement. Collocation points are sampled randomly over the domain with extra density near the airfoil.
- Case: low-Reynolds-number laminar flow (Re = 100) at α = 0°.

Potential-flow PINN (NACA 0024)

- Same input/output mapping, with 3 hidden layers of 100 neurons and tanh activation.
- Loss: PDE loss from continuity and irrotationality residuals, and a boundary loss that enforces the freestream velocity ($U_\infty \cos\alpha$, $U_\infty \sin\alpha$) in the far field; the angle of attack is 5°.

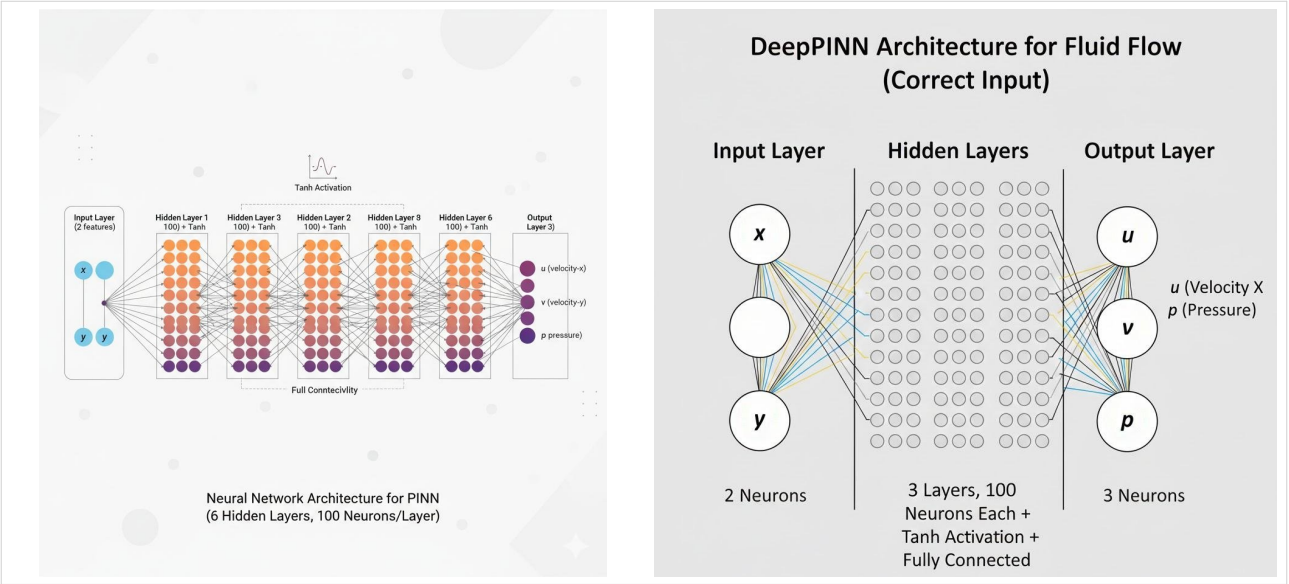


Figure 3. *PINN architectures: (a) Navier–Stokes PINN (6 × 100), (b) potential-flow PINN (3 × 100).*

4.3 Source Panel Method

- Potential flow with a source/sink on each panel; the strengths σ are found from the linear system A·σ = RHS that enforces zero normal velocity on the airfoil surface.
- The velocity and pressure fields are computed from σ and the panel geometry.
- Symmetric NACA 00xx airfoils are generated from the standard NACA thickness function with cosine spacing of the panels.
- Five thickness ratios t/c = 0.06, 0.12, 0.18, 0.24 and 0.30 are studied at α = 0°, with freestream speed 1 and density 1.

5. Results: CFD of the NACA 6409 Airfoil

5.1 Force coefficients at Re = 2×10⁶

Angle of attack	Cl (CFD)	Cd (CFD)
0°	0.7145	0.01315
5°	1.2474	0.01948

At 0° the experimental reference values used in the project are Cl ≈ 0.685 and Cd ≈ 0.01, giving a relative error of about 4.3 % in Cl and about 31 % in Cd. The drag is much more sensitive to the near-wall mesh resolution than the lift, which explains the larger drag error. At 5° the Cl value agrees with experiments within acceptable limits, while Cd again deviates because of its strong dependence on the mesh near the wall.

5.2 Flow fields

The pressure contours show a high-pressure stagnation region at the leading edge and a large suction (low-pressure) region on the upper surface; the velocity contours show the corresponding accelerated flow over the upper surface and a thin low-speed layer at the wall. When the angle of attack rises from 0° to 5° the suction on the upper surface and the leading-edge stagnation pressure increase, and the stagnation point moves toward the lower side of the leading edge. A wall-y^+ contour was also examined to check that the near-wall mesh is consistent with the wall treatment.

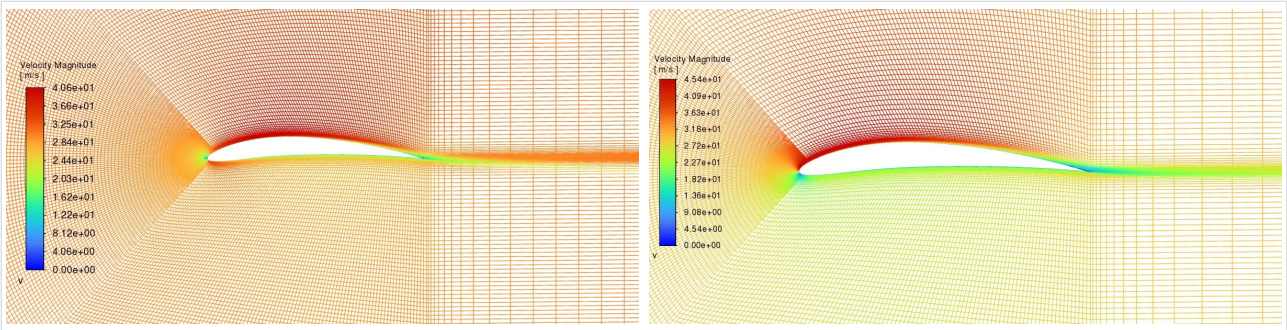


Figure 4. *Velocity magnitude contours at Re = 2×10⁶: (a) α = 0°, (b) α = 5°.*

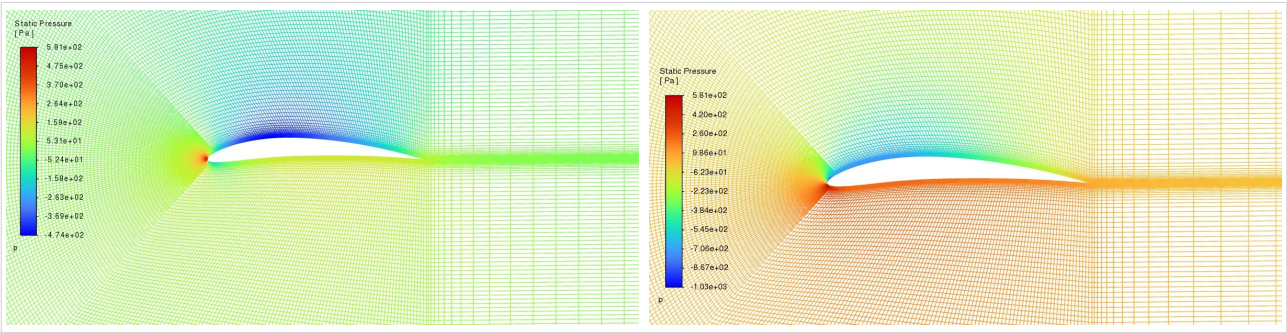


Figure 5. *Static pressure contours at Re = 2×10⁶: (a) α = 0°, (b) α = 5°.*

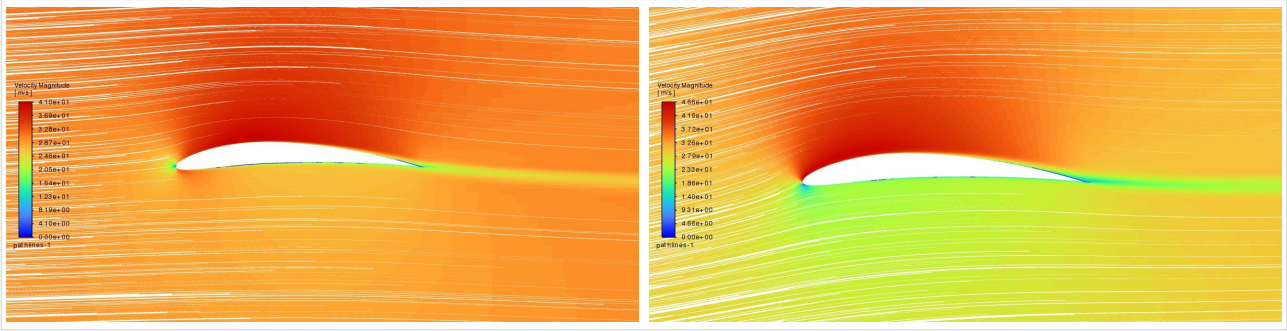


Figure 6. *Streamlines colored by velocity at Re = 2×10⁶: (a) α = 0°, (b) α = 5°.*

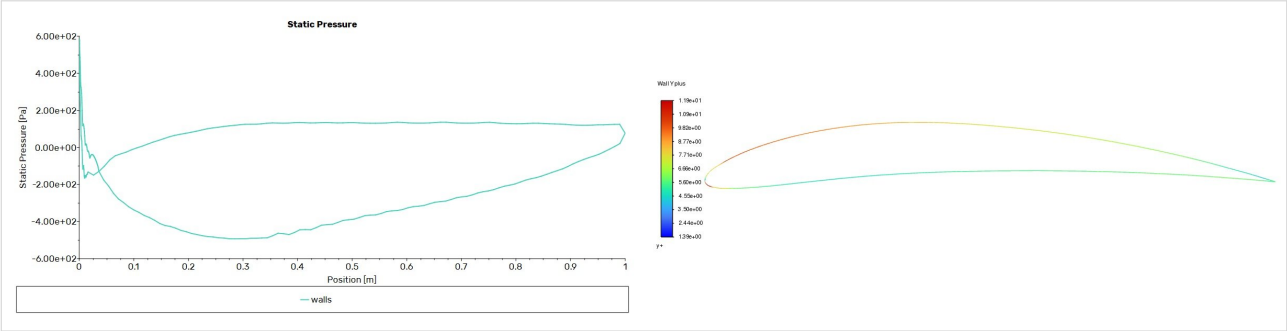


Figure 7. *(a) Static pressure along the airfoil surface at α = 0°; (b) wall y⁺ distribution.*

5.3 Mesh independence

Four meshes of different density were generated for Re = 3×10⁶ and α = 0°, changing only the number of elements and the resolution of the near-wall region; all other settings (Standard k–ε, boundary conditions) were kept the same.

Qualitative study: velocity contours, pressure contours, streamlines and surface-pressure distributions along the airfoil were compared for the four meshes. No noticeable change in the pressure distribution was found as the mesh was refined, so the solution is practically mesh-independent.

Quantitative study:

Mesh	Elements	Cl
Mesh 1 (very fine)	159,184	0.7195
Mesh 2 (fine)	97,574	0.7077
Mesh 3 (medium)	71,694	0.6681
Mesh 4 (coarse)	68,730	0.6794

The change in Cl decreases from the third to the first mesh. The Grid Convergence Index (GCI) of Cl on the finest mesh is about 0.87 %, so the finest mesh can be accepted as relatively mesh-independent, considering the balance between accuracy and cost.

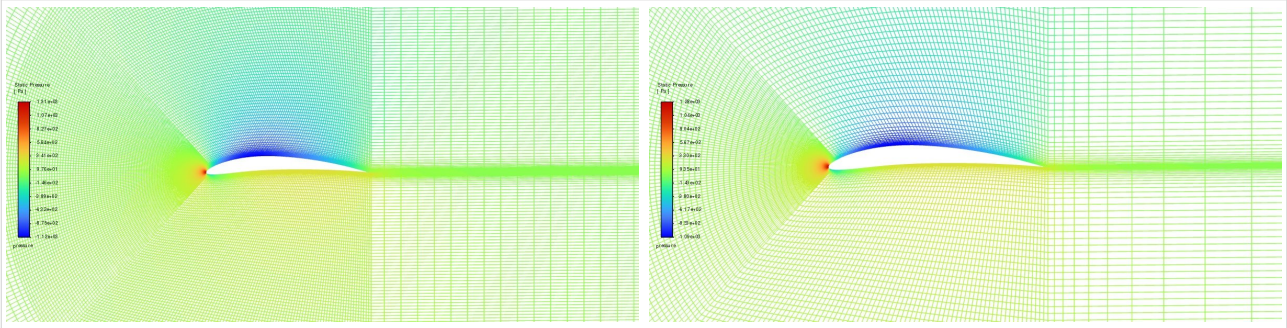


Figure 8. *Pressure contours at Re = 3×10⁶, α = 0°: (a) Mesh 1 (finest), (b) Mesh 4 (coarsest).*

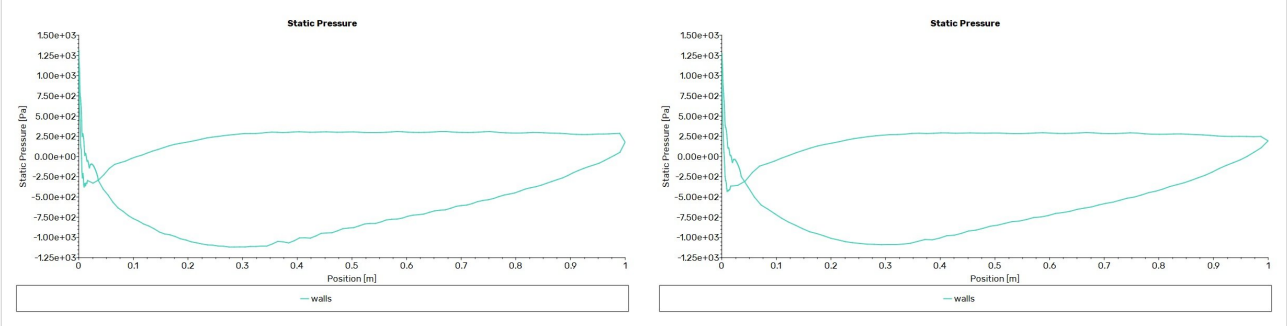


Figure 9. *Surface pressure along the airfoil: (a) Mesh 1, (b) Mesh 4. The distributions are practically identical.*

6. Results: Angle-of-Attack Study (NACA 6409, Re = 3×10⁶)

6.1 Contours and streamlines

- **α = −5°:** the stagnation point moves toward the upper half of the leading edge; the upper surface has relatively higher pressure and the lower surface lower pressure, which indicates negative or very small lift. The flow is fully attached.
- **α = 0°:** because of the camber, the upper surface already has lower pressure than the lower one, so the airfoil produces positive lift even at zero angle of attack. Velocity is highest near the leading edge and the thickest part on both sides, and streamlines are smooth and attached.
- **α = 15° (near stall):** the pressure difference between the upper and lower surfaces grows sharply, with very strong suction on the upper surface near the leading edge. Toward the trailing edge the streamlines no longer follow the surface and a large wake/separated region appears, showing that the airfoil is close to stall or beyond it.

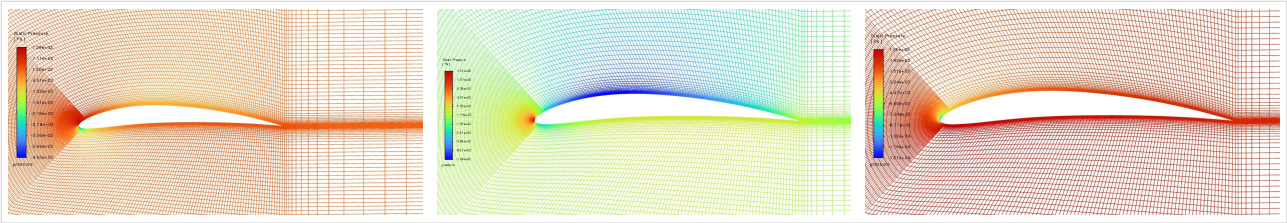


Figure 10. *Static pressure contours: (a) α = −5°, (b) α = 0°, (c) α = 15°.*

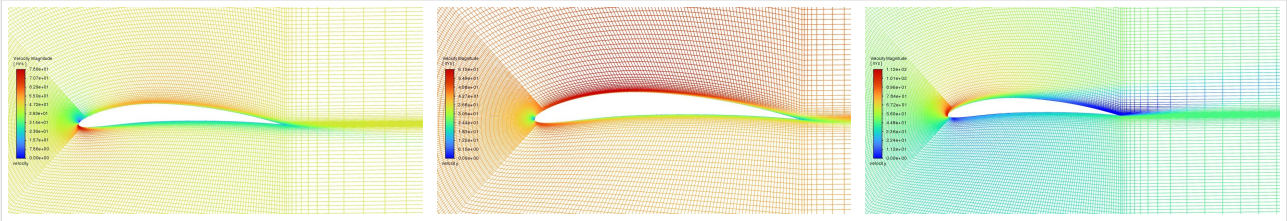


Figure 11. *Velocity magnitude contours: (a) α = −5°, (b) α = 0°, (c) α = 15°.*

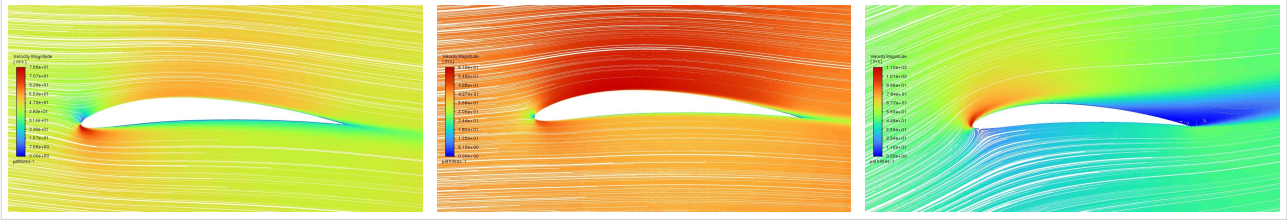


Figure 12. *Streamlines: (a) α = −5°, (b) α = 0°, (c) α = 15° (near stall).*

6.2 Lift curve

The lift coefficient increases linearly with angle of attack from about −5° to about 10°, where CFD and experimental curves agree closely; the lift at 0° is about 0.7, confirming that the cambered NACA 6409 produces lift without an installation angle. Maximum lift occurs near 15°, after which the airfoil enters stall and Cl decreases because of strong flow separation on the upper surface. In the post-stall region CFD predicts a sharper loss of lift than the experiment, a typical weakness of RANS turbulence models in separated flow.

Parameter	Experimental	CFD
Zero-lift angle	−7°	−6.5°
Lift-curve slope in the linear region (per degree)	0.0867	0.1067
Maximum lift coefficient	1.63	1.63
Stall angle	15°	15°

The zero-lift angles differ by only 0.5°, which is acceptable and can be attributed to numerical simplifications, the two-dimensional assumption and differences between the boundary conditions and the wind-tunnel conditions. The larger CFD slope means that CFD tends to over-predict lift at intermediate angles, for the usual reasons: idealized modeling of the flow, limitations of the turbulence model and the missing three-dimensional effects. The maximum lift and the stall angle agree very well.

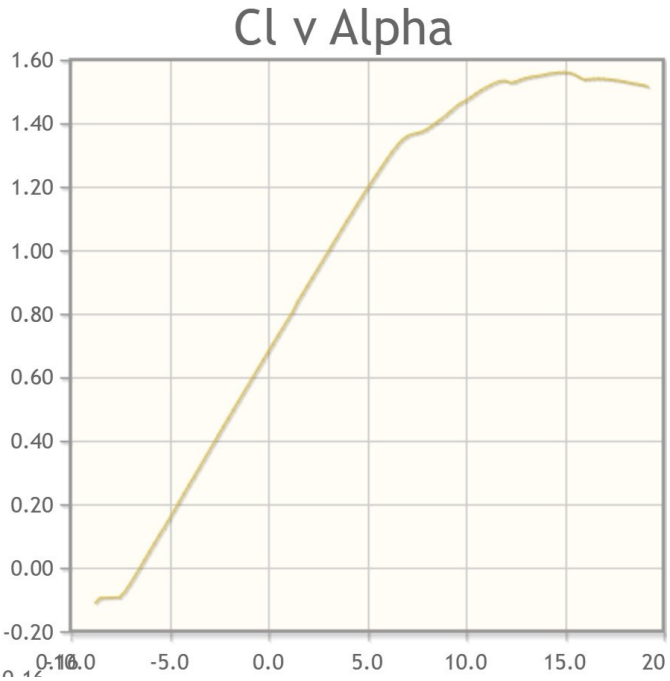


Figure 13. *Reference (experimental) lift coefficient versus angle of attack for NACA 6409.*

6.3 Drag curve and lift-to-drag ratio

The experimental drag coefficient is small and almost constant over the studied range, as expected for an efficient airfoil. The CFD drag agrees with the experiment only at very low angles (−5° to 0°). At 10° and 15° the computed Cd becomes **negative** (roughly −0.25 and −0.36 in the plotted curve), which is physically impossible for drag. The report treats this as a computational error in the CFD drag calculation; the orientation of the force axes was checked and was not the cause, and the likely remedy is a more refined mesh behind the airfoil.

As a consequence, the CFD lift-to-drag ratio is strongly under-estimated: the experimental curve peaks at about 85 near 0° and falls to about 69 at 10° and about 26 at 15° (close to stall), whereas the CFD curve peaks at about 58 at 0° and drops to only about 5–6 at 10° and 15°. The lift calculation is reliable, but the drag results at high angle of attack should not be used quantitatively.

6.4 Pressure coefficient

The negative pressure coefficient (−Cp) on the upper surface grows steadily with angle of attack, with a clear jump at 10° and 15°, showing the strong acceleration on the upper surface that dominates lift generation. On the lower surface −Cp is mostly negative and becomes more negative with angle of attack, i.e. the pressure there rises, with the largest lower-surface pressure at 15°. The pressure difference between the two surfaces therefore increases with angle of attack and is largest at 15°, which is consistent with the approach to stall (strong suction, but possible onset of separation).

6.5 Solver convergence

Scaled residual histories were recorded: 300 iterations for the viscous k–ε case and 200 iterations for the inviscid case, with continuity and velocity residuals decreasing to very low levels.

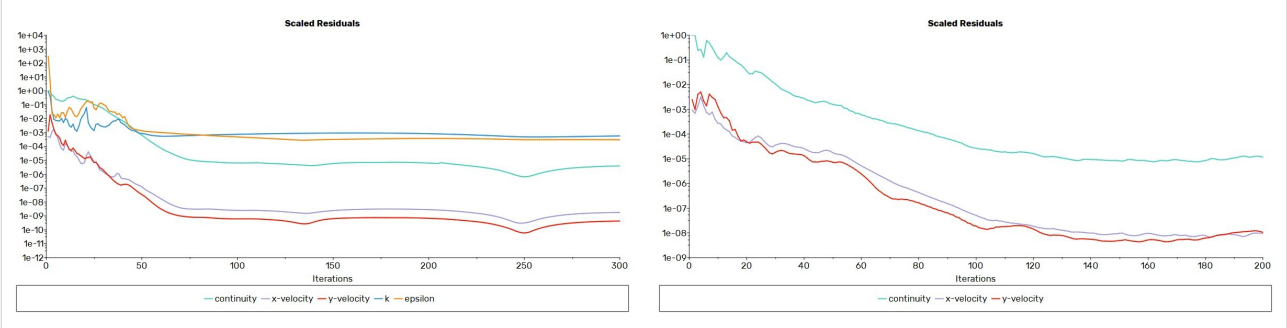


Figure 14. *Scaled residual histories: (a) viscous k–ε case (NACA 6409), (b) inviscid case (NACA 0024).*

7. Results: Inviscid CFD of NACA 0024 (α = 5°)

The flow is potential, so boundary-layer growth, separation and stall are not modeled. The velocity contour shows increased speed on the upper surface (lowest pressure, the main source of lift), slower flow on the lower surface, and a small stagnation region near the leading edge on the lower side. The static-pressure contour mirrors this: low pressure on the upper surface, higher pressure on the lower surface, and maximum pressure at the stagnation point. The streamlines follow the body without separation. These results are used as a reference for the potential-flow PINN.

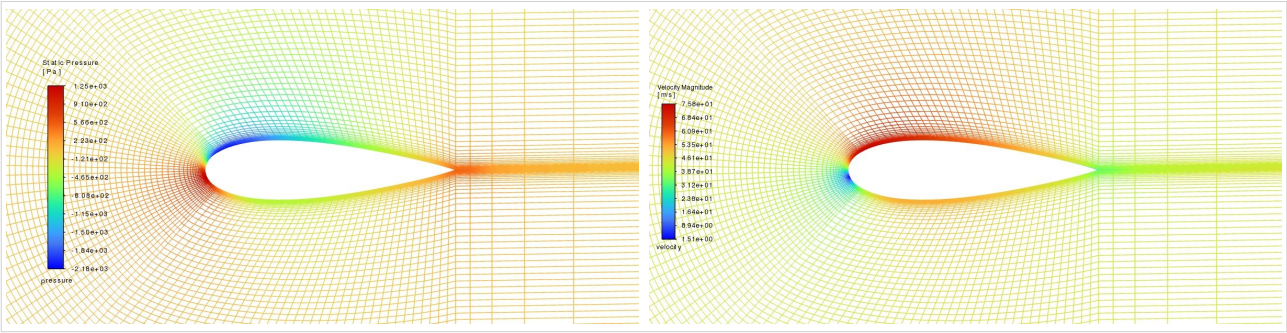


Figure 15. *Inviscid CFD, NACA 0024 at α = 5°: (a) static pressure, (b) velocity magnitude.*

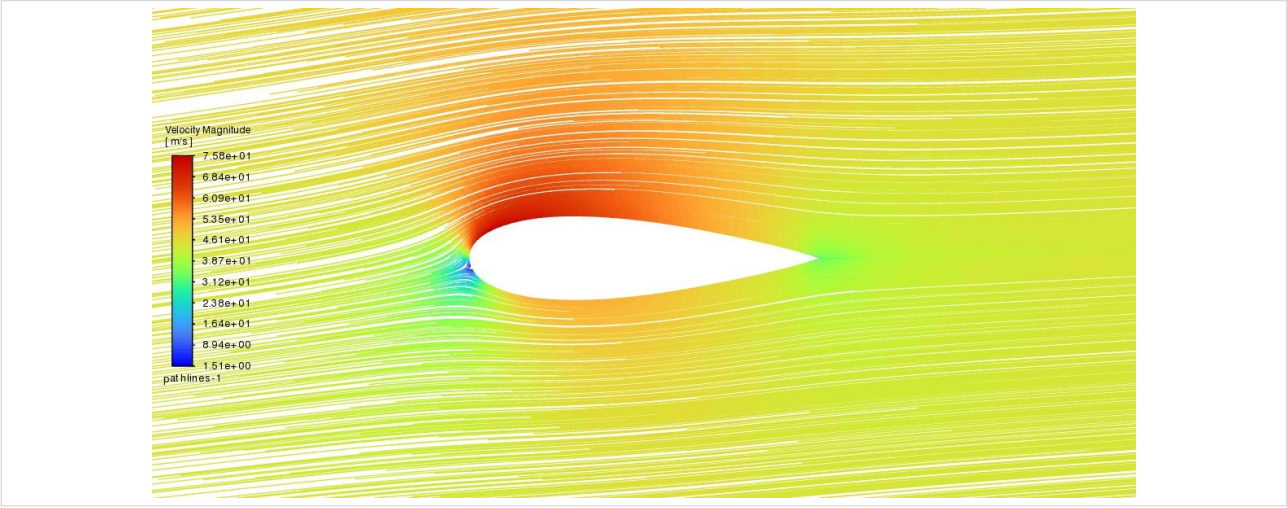


Figure 16. *Inviscid CFD, NACA 0024 at α = 5°: streamlines colored by velocity.*

8. Results: Navier–Stokes PINN (NACA 6409, α = 0°)

8.1 Training

The loss history shows the total loss decreasing during training; the boundary-condition loss decreases quickly at the beginning (it is weighted more heavily) while the PDE loss stays lower, and the L-BFGS phases effectively reduce the total loss in every cycle. The vertical lines mark the L-BFGS phases, and the jumps in the curve occur because intermediate L-BFGS iterations are not logged. At the end all loss terms reach low values (of the order 10^{-2} to 10^{-3}).

Depth and width study. To check whether the network size matters, the code was run for several numbers of hidden layers and neurons per layer. The final total loss was of the same order for all combinations (about 10^{-2} to 10^{-3}), so the solution is not significantly sensitive to the exact depth or width.

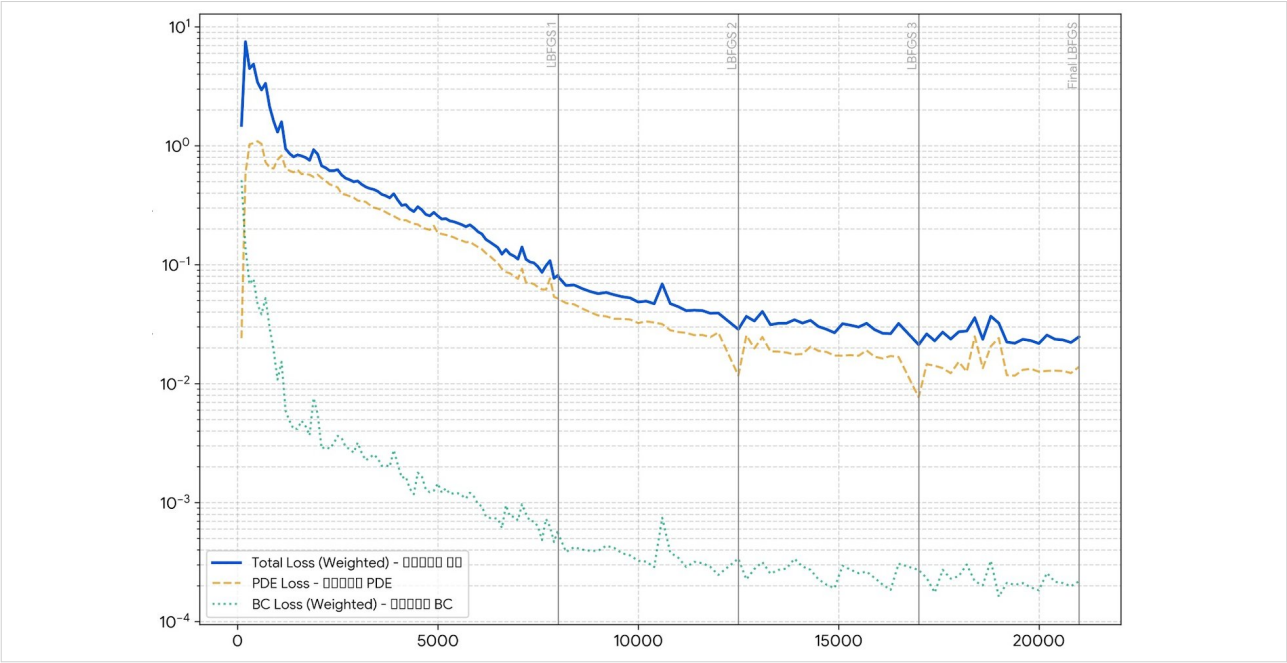


Figure 17. *Training loss history of the Navier–Stokes PINN (total, PDE and boundary-condition losses; vertical lines mark L-BFGS phases).*

8.2 Flow field

- **Pressure:** a high-pressure stagnation region at the leading edge, a large suction region over the upper part of the airfoil, and some oscillation or local peak near the trailing edge.
- **Velocity:** accelerated flow over the upper surface consistent with Bernoulli's principle, a low-speed layer near the wall (no-slip), nearly uniform freestream far from the airfoil, and some local velocity peaks near the leading and trailing edges that may be numerical artifacts.
- **Streamlines:** flow is mostly attached with a small wake near the trailing edge and approximately symmetric behavior at 0°.
- **Surface pressure:** a positive peak at the leading edge (stagnation point) and a pronounced dip near 45–50 % of the chord, where the pressure difference between the two surfaces is largest; this location matches the position of maximum thickness of the airfoil, which indicates that the simulation is consistent with the physics of the problem.

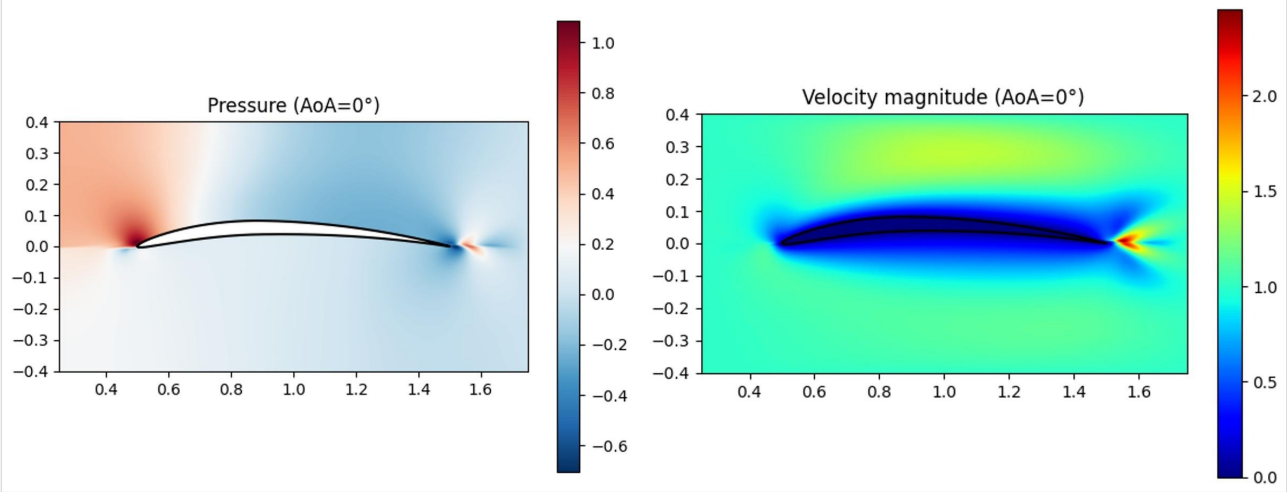


Figure 18. *Navier–Stokes PINN, α = 0°: (a) pressure field, (b) velocity magnitude.*

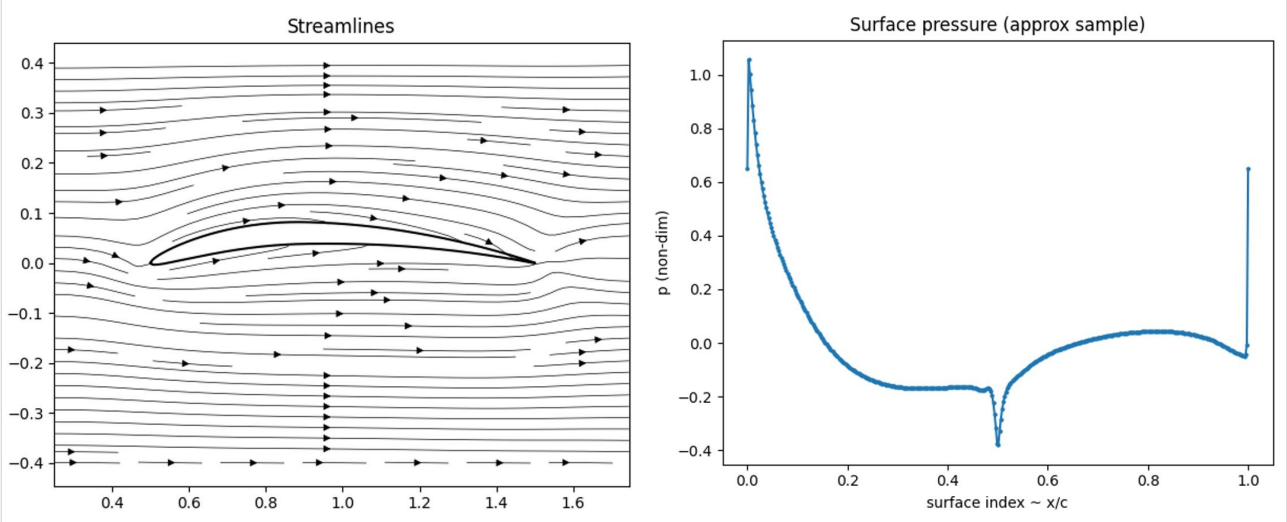


Figure 19. *Navier–Stokes PINN, α = 0°: (a) streamlines, (b) surface pressure distribution.*

8.3 Force coefficients

For Re = 100 and α = 0° the PINN gives Cl = 0.05603 and Cd = 0.05203. No experimental data exist for Re = 100, so the lowest available reference Reynolds number (Re = 50,000) was used, with Cl = 0.3 and Cd = 0.048. The relative errors are about 81 % for Cl and 8.4 % for Cd. These numbers should be read with care because the two Reynolds numbers differ by three orders of magnitude.

8.4 Discussion of discrepancies

- Very sharp peaks and small oscillations, especially near the leading and trailing edges, are mostly numerical: PINNs learn steep gradients with difficulty.
- Insufficient boundary sampling near the leading and trailing edges leads to weak approximation of the surface pressure.
- Imbalance of the loss weights (for example weakly weighted pressure or velocity boundary terms) reduces the accuracy of the surface pressure.
- Geometry/mesh representation errors: describing the airfoil boundary with few or imprecise points introduces local errors.

9. Results: Potential-Flow PINN (NACA 0024, α = 5°)

- **Pressure contour:** a high-pressure stagnation point at the leading edge, slightly displaced toward the lower side as expected for 5°; a large low-pressure region on the upper surface and higher pressure below, whose difference generates lift. The color transitions are smooth, and there is no wake or viscous effect behind the airfoil, as appropriate for potential flow.
- **Velocity contour:** speed exceeds the freestream over the upper and lower body (maximum around 1.2–1.4 times U∞), is near zero at the leading-edge stagnation point, and returns to the freestream value in the far field. High velocity corresponds to low pressure, so the network respects Bernoulli's relation. No boundary layer is present, confirming that slip-flow physics was learned.
- **Streamlines:** smooth, parallel in the far field, curved around the body, denser on the upper side, and returning to nearly parallel behind the trailing edge without vortices or recirculation.
- **Training:** the loss history shows rapid initial decrease followed by slow convergence over about 5,000 iterations, with the boundary loss dominating.

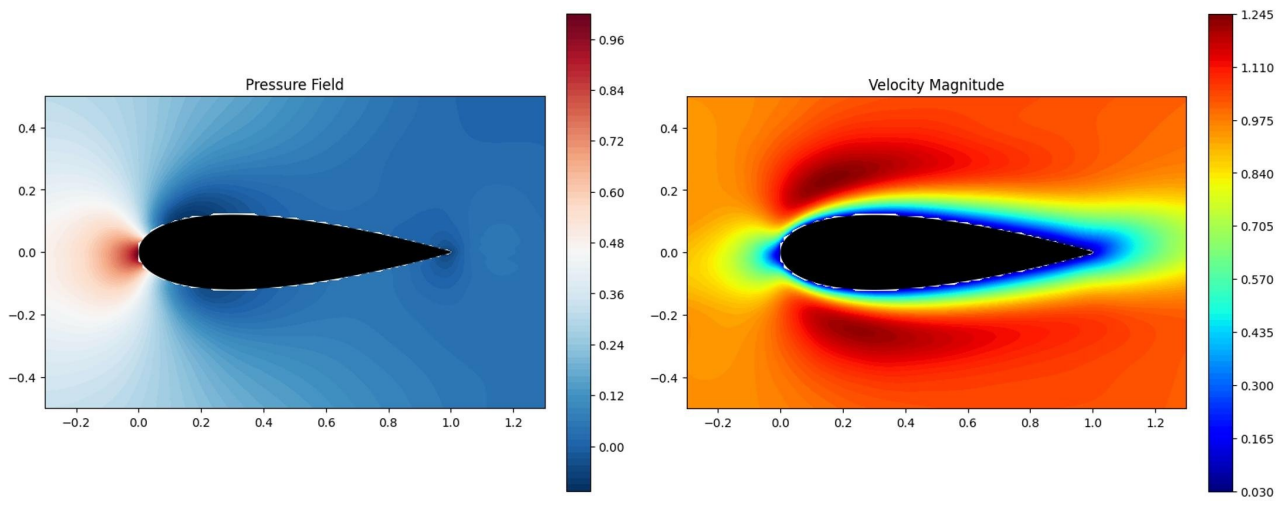


Figure 20. *Potential-flow PINN, NACA 0024 at α = 5°: (a) pressure field, (b) velocity magnitude.*

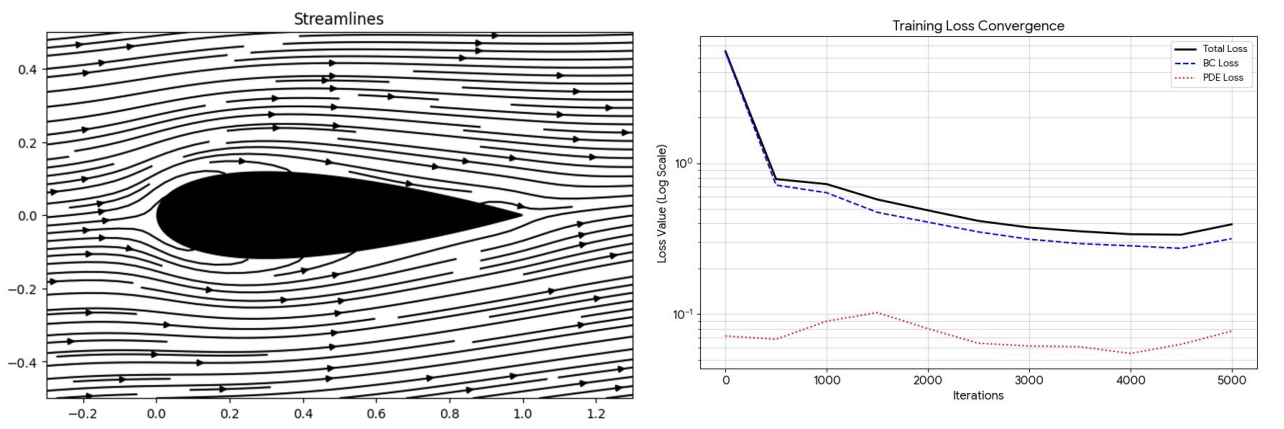


Figure 21. *Potential-flow PINN: (a) streamlines, (b) training loss convergence (total, BC and PDE losses).*

10. Results: Source Panel Method (NACA 00xx, α = 0°)

Panel independence. Because each panel carries one source/sink, the panel-independence study is effectively a convergence study with respect to the number of source/sink elements. With few panels the numerical drag coefficient is relatively large because of discretization error and the large spacing between sources; it drops rapidly as the number of panels increases and then settles to an almost constant value, so the numerical solution is stable with respect to the panel count.

Effect of thickness. The numerical Cd decreases as the thickness ratio t/c increases (in the plotted study from about 5.7 at t/c = 0.06 to about 0.5 at t/c = 0.30). Thicker airfoils give a smoother pressure distribution and smaller pressure differences between neighboring panels, whereas for thin airfoils the panels are very close together and numerical fluctuations in σ and p are larger.

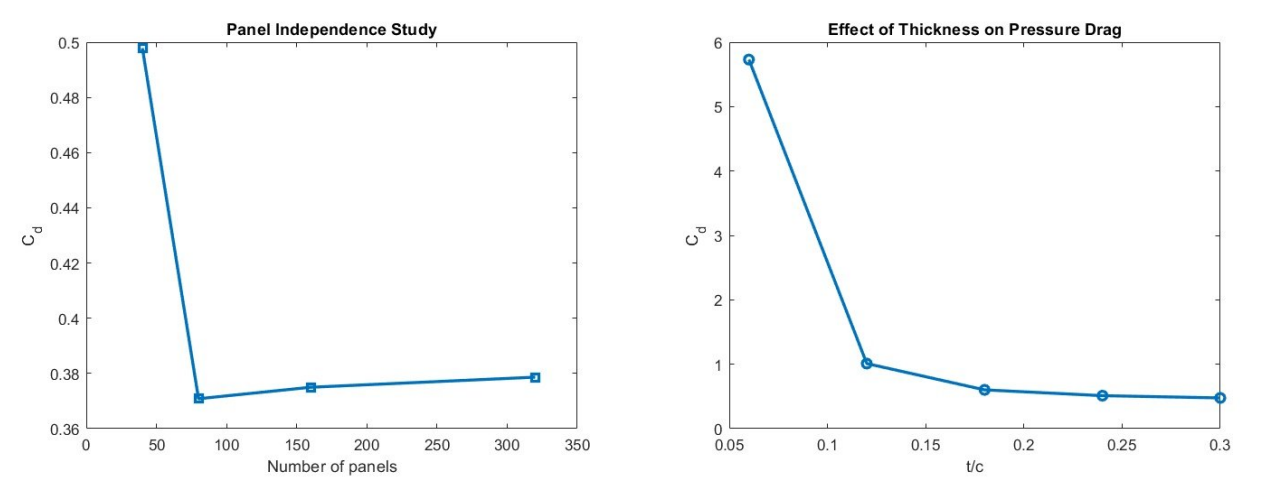


Figure 22. *Panel method: (a) panel independence study, (b) effect of thickness ratio t/c on the numerical pressure drag.*

Streamlines and equipotential lines. For the thickest airfoil (t/c = 0.30) the ψ contours are steady and symmetric about the chord; far from the airfoil they are nearly horizontal and parallel to the freestream, which confirms that the far-field boundary condition is applied correctly. Near the leading and trailing edges and in regions of rapid geometric change, small discontinuities appear in the contours; they are not real separation but numerical effects of the panel method and the plotting grid, and they decrease with finer panels. The φ lines are nearly perpendicular to the freestream far away, bend toward the leading and trailing edges near the body, are perpendicular to the ψ lines, and become denser and more curved with increasing thickness while symmetry is preserved.

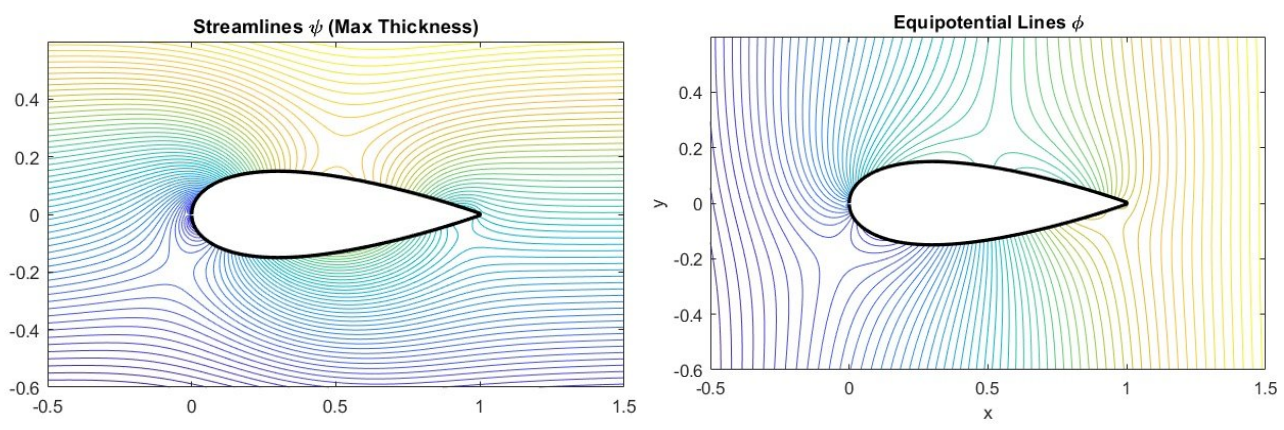


Figure 23. *Panel method, t/c = 0.30: (a) streamlines (ψ), (b) equipotential lines (φ).*

Validation. The results were checked against potential-flow theory: symmetry of the pressure distribution at zero angle of attack, steady flow with no vortices or separation, and continuity of the far field. According to d'Alembert's paradox the true drag in ideal flow is zero, so any non-zero

numerical Cd reflects thickness and panel discretization rather than real drag. The Cp distribution computed from the panel strengths was compared with the Joukowski airfoil behavior for a symmetric section at zero angle of attack and agrees at least qualitatively.

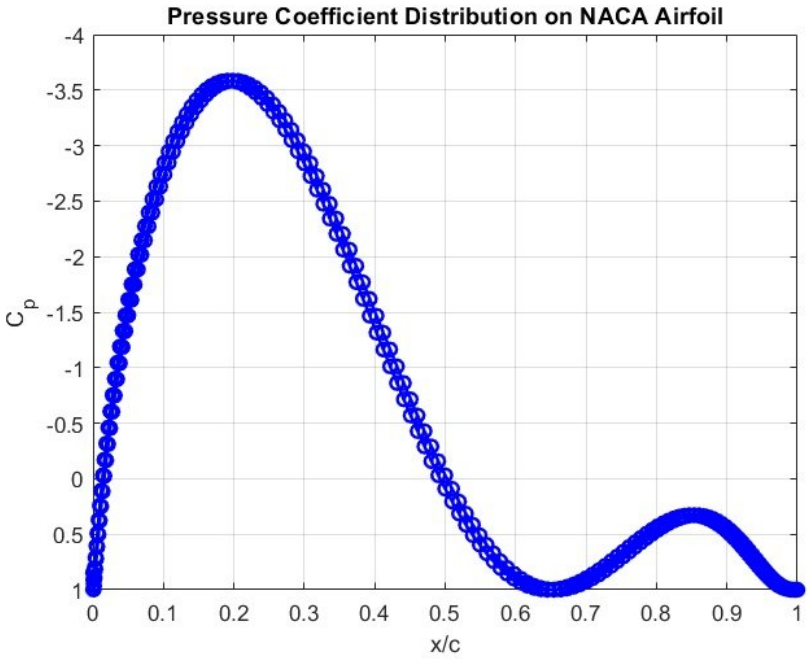


Figure 24. *Pressure coefficient (Cp) distribution on a symmetric NACA airfoil from the panel method.*

11. Comparison of the Three Approaches

Aspect	Panel method	CFD (Fluent)	PINN
Physics	Inviscid, incompressible potential flow	Viscous RANS (k–ε)	Navier–Stokes (laminar, low Re) or potential flow
Mesh	Surface panels only	Volume mesh (C-type)	Mesh-free (collocation points)
Cost	Very low, near-instant	High	High training cost, then cheap evaluation
Viscous effects, separation, stall	Not captured	Captured (with turbulence-model limits)	Not captured in this study's low-Re/potential cases
Lift	Qualitative	Good (4.3 % error at 0°, matching C_l,max and stall angle)	Qualitative
Drag	Not physical (zero in theory)	Good at low angles, unreliable at high angles in this study	Strong dependence on Re and setup
Main weaknesses	No drag, no separation	Cost; drag sensitivity to near-wall mesh	Loss-weight sensitivity, steep-gradient accuracy, training cost

12. Conclusions

1. The source panel method gives a fast and approximate pressure distribution and streamlines at very low computational cost. It correctly shows symmetry at zero angle of attack and the effect of thickness on the flow and pressure distribution, and its results converge as the number of panels increases and agree with potential-flow theory, although some small discontinuities from discretization remain.
2. Fluent simulations solving the Navier–Stokes equations provide the flow and pressure fields in high detail and allow examination of the lift and drag forces and of the regions of real flow separation. For NACA 6409 the lift curve, zero-lift angle, maximum lift (1.63) and stall angle (15°) agree well with experimental data, while drag at high angles of attack is not reliable and needs a more refined mesh behind the airfoil.
3. The PINN approach solved the governing equations and reproduced the flow field without explicit meshing, showing the capability of physics-based, data-driven methods for complex flows. Its accuracy, however, depends on the training setup, and its cost and need for reference data remain significant.
4. Overall: the panel method is suitable for quick, approximate analysis of ideal flow; CFD has the capability to give detailed flow information and accurate force calculation; and PINNs are a new method that enables mesh-free flow reconstruction with acceptable accuracy and has a promising future for intelligent applications in computational fluid dynamics.

13. Limitations and Notes on the Data

- The Navier–Stokes PINN was run at Re = 100 (laminar), while the Fluent cases use Re = 2–3×10^6 with a turbulence model; the PINN force coefficients were compared with data for Re = 50,000, so the comparison is only qualitative.
- The negative CFD drag at 10° and 15° is an unresolved computational problem and affects the CFD lift-to-drag curve.
- The numerical drag from the panel method is not zero and can be large for thin airfoils (for example about 5.7 at t/c = 0.06); it should be interpreted as a discretization effect, not as physical drag.
- The qualitative discussion of the lift curve in the original report quotes maximum lift values of 1.65 (CFD) and 1.57 (experiment), whereas the comparison table gives 1.63 for both; the table values are used in this report.

14. Source Code and Appendix

The original report contains the full Python PINN codes (Navier–Stokes case and potential-flow case), the MATLAB source panel method code, the residual-convergence plots of the Fluent runs, and the complete set of contour, streamline and Cp figures referred to above.